# Lecture 4: Practice Problems

Math for Machine Learning

November 14, 2026

## 1 Probability Spaces

**Problem 1.1** (Mandatory)**.** Two fair dice are rolled. Find the probability that

(a) the sum is 8;

(b) the two dice show the same number;

(c) the sum is at least 10;

(d) at least one die shows a 3;

(e) the larger of the two numbers is 4;

(f) the sum is even, given that the first die shows a 3.

**Problem 1.2** (Mandatory)**.** A factory has three machines, which produce 50%, 30% and 20% of all items. Their defect rates are 1%, 2% and 3%.

(a) What fraction of all items is defective?

(b) An item is defective. What is the probability that it was produced by the third machine? By the second?

**Problem 1.3** ($\star$)**.** Sofia and Tess each choose one of the integers $1, 2, \ldots, 10$ uniformly at random, independently of each other. What is the probability that neither of the two numbers is the square of the other?

**Problem 1.4** ($\star$)**.** An integer is chosen uniformly at random from $1, 2, \ldots, 1000$. Use the inclusion–exclusion formula to compute the probability that it is divisible by 3 or by 5, and the probability that it is divisible by 3, 5 or 7.

**Problem 1.5** ($\star$)**.** A fair die is rolled until the first six appears. What is the probability that this happens at an even-numbered roll? (Hint: the event is a countable disjoint union, or condition on the first roll.)

**Problem 1.6.** A fair die is rolled. Let $A =$ "the result is even", $B =$ "the result is at most 4", and $C =$ "the result is at most 3". Which of the pairs $(A, B)$, $(A, C)$ and $(B, C)$ are independent?

**Problem 1.7** ($\star$)**.** (The Monty Hall problem.) A car is hidden behind one of three doors, each with probability $\frac{1}{3}$. You choose door 1. The host, who knows where the car is, opens one of the two other doors that does not hide the car (choosing uniformly at random if both are possible), say door 3. Use Bayes' theorem to compute the probability that the car is behind door 2. Should you switch?

## 2 Random Variables and Their Distributions

**Problem 2.1** (**Mandatory**)**.** A student guesses the answers to 20 multiple-choice questions, each with 4 options, independently and uniformly at random. Let $X$ be the number of correct answers.

(a) What is the distribution of $X$? Compute $P(X = 5)$ and $P(X \geq 1)$.

(b) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

**Problem 2.2.** The number of emails that arrive in an hour is Poisson(3). Compute the probability that no email arrives, that at least two emails arrive, and that exactly three emails arrive.

**Problem 2.3** (**Mandatory**)**.** Let $X$ have the density $p(x) = cx^2$ for $0 \leq x \leq 2$, and $p(x) = 0$ otherwise.

(a) Find $c$ and the CDF of $X$.

(b) Compute $P(X \leq 1)$ and the median of $X$, i.e., the number $m$ with $P(X \leq m) = \frac{1}{2}$.

**Problem 2.4.** Let $U \sim \mathrm{Uniform}(0, 1)$. Find the densities of (a) $\sqrt{U}$; (b) $\frac{1}{U}$; (c) $-2 \log U$. Which of them is a distribution you know?

**Problem 2.5** ($\star$)**.** Use inverse transform sampling to find a formula that turns a uniform random number $U \in (0, 1)$ into a sample of

(a) the distribution with the density $2x$ on $[0, 1]$;

(b) the Pareto distribution with the CDF $F(x) = 1 - x^{-a}$ for $x \geq 1$ (with $a > 0$), a model for incomes and file sizes;

(c) the Cauchy distribution with the CDF $F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x$.

**Problem 2.6.** Let $X$ take the values 0, 1 and 3 with the probabilities 0.2, 0.5 and 0.3.

(a) Sketch the CDF $F$ of $X$, and check that $P(X = 1)$ is the jump $F(1) - \lim_{y \uparrow 1} F(y)$.

(b) Compute the quantile function $F^{-1}(u) = \min\{x : F(x) \geq u\}$ at $u = 0.1$, 0.2, 0.65, 0.7 and 0.71.

(c) Describe how inverse transform sampling turns a uniform random number $U$ into a sample of $X$.

**Problem 2.7.** The heights of adults in a population are approximately $\mathcal{N}(170, 10^2)$ (in cm). What fraction of the people are between 160 and 190 cm tall? What fraction is taller than 185 cm? Find an interval around 170 that contains 95% of the population. (Use $\Phi(1) \approx 0.8413$, $\Phi(1.5) \approx 0.9332$, $\Phi(2) \approx 0.9772$.)

## 3 Expectation and Variance

**Problem 3.1** (**Mandatory**)**.** (a) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ for the random variable of Problem 2.3.

(b) Let $X$ be uniform on $\{1, 2, \ldots, n\}$. Show that $\mathbb{E}[X] = \frac{n+1}{2}$ and $\mathrm{Var}(X) = \frac{n^2-1}{12}$.

**Problem 3.2** ($\star$)**.** The numbers $1, \dots, n$ are shuffled into a uniformly random order. Let $X$ be the number of fixed points, i.e., of numbers that stay at their original position. Show that $\mathbb{E}[X] = 1$ for every $n$. (Hint: write $X$ as a sum of indicator variables, and use linearity of expectation.)

**Problem 3.3.** A random variable $X \geq 0$ has mean 10.

(a) Bound $P(X \geq 50)$.

(b) If moreover $\mathrm{Var}(X) = 4$, bound $P(|X - 10| \geq 5)$, and $P(X \geq 15)$.

**Problem 3.4.** (a) Let $X$ be uniform on $\{1, 4\}$. Compute $\mathbb{E}[\sqrt{X}]$ and $\sqrt{\mathbb{E}[X]}$, and explain the inequality between them with Jensen's inequality.

(b) Show that $\mathbb{E}[X^4] \geq \mathbb{E}[X^2]^2$ for every random variable $X$.

**Problem 3.5** ($\star$)**.** (The coupon collector.) Each box of cereal contains one of 6 different toys, each with probability $\frac{1}{6}$, independently. Show that the expected number of boxes needed to collect all 6 toys is $6(1 + \frac{1}{2} + \cdots + \frac{1}{6}) = 14.7$. (Hint: after you have $k$ different toys, the number of boxes until a new toy appears is Geometric($\frac{6-k}{6}$).)

## 4 Joint Distributions and Independence

**Problem 4.1** (**Mandatory**)**.** Let $X \in \{0, 1, 2\}$ and $Y \in \{0, 1\}$ have the joint pmf

| | $Y = 0$ | $Y = 1$ |
|---|---|---|
| $X = 0$ | 0.1 | 0.1 |
| $X = 1$ | 0.2 | 0.1 |
| $X = 2$ | 0.1 | 0.4 |

(a) Find the marginal distributions, and the conditional probabilities $P(Y = 1 \mid X = x)$ for $x = 0, 1, 2$. Are $X$ and $Y$ independent?

(b) Compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathrm{Cov}(X, Y)$ and $\mathrm{Corr}(X, Y)$.

(c) Compute $\mathbb{E}[X \mid Y = 0]$ and $\mathbb{E}[X \mid Y = 1]$, and check the tower property.

**Problem 4.2** (**Mandatory**)**.** Let $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$ and $\mathrm{Cov}(X, Y) = 3$.

(a) Compute $\mathrm{Var}(X + Y)$, $\mathrm{Var}(2X - Y)$ and $\mathrm{Corr}(X, Y)$.

(b) (Ensembles and portfolios.) Find the $t \in \mathbb{R}$ that minimizes $\mathrm{Var}(tX + (1 - t)Y)$, and the minimal variance. Compare it with $\mathrm{Var}(X)$.

**Problem 4.3.** Let $(X, Y)$ have the joint density $p(x, y) = x + y$ on the unit square $[0, 1]^2$.

(a) Check that $p$ is a density, and find the marginal density of $X$.

(b) Compute $\mathbb{E}[X]$ and $\mathrm{Cov}(X, Y)$.

(c) Find the conditional density $p(y \mid x)$ and $\mathbb{E}[Y \mid X = x]$.

**Problem 4.4.** (a) Let $X$ and $Y$ be independent and Uniform$(0, 1)$. Compute the density of $X + Y$ as a convolution, and use it to compute $P(X + Y \leq 1.5)$.

(b) The numbers of emails that arrive in an hour from two independent sources are Poisson(2) and Poisson(3). What is the distribution of the total number, and what is the probability that no email arrives?

**Problem 4.5** ($\star$)**.** Let $(X, Y)$ be uniformly distributed on the four points $(1, 0)$, $(-1, 0)$, $(0, 1)$, $(0, -1)$. Show that $X$ and $Y$ are uncorrelated, but not independent.

**Problem 4.6.** A fully connected layer computes $z_i = \sum_{j=1}^{400} W_{ij} x_j$ from 400 inputs with $\mathbb{E}[x_j^2] = 1$, with independent weights $W_{ij} \sim \mathcal{N}(0, v)$ that are independent of the inputs.

(a) Find $\mathrm{Var}(z_i)$. For which $v$ is it equal to 1?

(b) The outputs pass through a ReLU. Find $\mathbb{E}[\mathrm{ReLU}(z_i)^2]$ for $v = \frac{1}{400}$ and for $v = \frac{2}{400}$, and the second moment of the outputs after 20 such layers (of width 400) in both cases.

(c) For weights that are uniform on $[-b, b]$, which $b$ gives the same variance as $v = \frac{2}{400}$?

**Problem 4.7.** For the random variables of Problem 4.1, find the best predictor $m(X)$ of $Y$ in the sense of the mean squared error, and its mean squared error $\mathbb{E}[(Y - m(X))^2]$. Compare with the mean squared error of the best constant prediction.

**Problem 4.8** (**Mandatory**)**.** (a) Find the moment generating function of the number $X$ shown by a fair die.

(b) Let $X \sim \mathrm{Exponential}(\lambda)$, with $M_X(t) = \frac{\lambda}{\lambda - t}$ for $t < \lambda$. Use the cumulant generating function to find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

(c) Find the moment generating function of the sum of $n$ independent $\mathrm{Exponential}(\lambda)$ variables, and its mean and variance.

**Problem 4.9** ($\star$)**.** Let $X \sim \mathrm{Poisson}(\lambda)$ and $a > \lambda$. Show that the optimized Chernoff bound is

$$P(X \geq a) \leq \exp\Big( - \Big( a \log \frac{a}{\lambda} - a + \lambda \Big) \Big).$$

Evaluate it for $\lambda = 10$ and $a = 20$, and compare with Chebyshev's inequality and with the exact value $P(X \geq 20) \approx 0.0035$.

**Problem 4.10.** Emails arrive as a Poisson process with the rate 3 per hour.

(a) What is the probability that no email arrives in 20 minutes?

(b) What is the probability of at least 2 emails in one hour?

(c) What is the expected time until the 5th email, and what is the density of the arrival time of the 2nd email?

(d) No email has arrived in the first 30 minutes. What is the probability that none arrives in the next 20 minutes?

**Problem 4.11.** (a) One of two coins is chosen with probability $\frac{1}{2}$ each; their head probabilities are 0.3 and 0.9. The chosen coin is flipped twice, with the results $X$ and $Y$ (1 for heads). Show that $X \perp\!\!\!\perp Y \mid Z$, where $Z$ is the chosen coin, and compute $P(X = 1)$, $P(X = 1, Y = 1)$ and $P(Y = 1 \mid X = 1)$. Are $X$ and $Y$ independent?

(b) A burglary ($B = 1$, probability 0.01) and an earthquake ($E = 1$, probability 0.02) occur independently, and the alarm goes off ($A = 1$) exactly when at least one of them occurs. Compute $P(B = 1 \mid A = 1)$ and $P(B = 1 \mid A = 1, E = 1)$, and explain the difference.

**Problem 4.12** (Mandatory)**.** A naive Bayes spam filter uses $P(\text{spam}) = 0.3$ and two words: "winner" appears in 20% of spam and 1% of other emails, and "report" in 5% of spam and 25% of other emails.

(a) Compute $P(\text{spam} \mid \mathbf{x})$ for an email that contains both words, and for one that contains "winner" but not "report".

(b) Write the log-odds $\log \frac{P(\text{spam}|\mathbf{x})}{P(\text{other}|\mathbf{x})}$ as an affine function $w_0 + w_1 x_1 + w_2 x_2$ of the indicators $x_1$ ("winner") and $x_2$ ("report").

(c) Which assumption makes the classifier "naive"? Give a reason why it may fail for words in emails.

## 5 Random Vectors and Covariance Matrices

**Problem 5.1** (Mandatory)**.** Let $\mathbf{X}$ have mean $\mathbf{0}$ and covariance matrix $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

(a) Compute $\operatorname{Var}(X_1 + X_2)$ and $\operatorname{Var}(X_1 - X_2)$. In which direction is the variance largest?

(b) Compute the whitening matrix $\Sigma^{-1/2}$, and check that $\Sigma^{-1/2}\mathbf{X}$ has covariance matrix $I$.

(c) Compute the Mahalanobis distances of $(1, 1)$ and of $(1, -1)$ from $\mathbf{0}$. Why are they different, although both points have the same Euclidean norm?

**Problem 5.2** (Mandatory)**.** For the data points $(0, 1)$, $(1, 3)$, $(2, 2)$, $(3, 6)$, compute the sample mean, the sample covariance matrix (with the factor $\frac{1}{N}$ and with $\frac{1}{N-1}$), the sample correlation, and the direction of the first principal component.

**Problem 5.3.** Let $\mathbf{X}$ have mean $(1, -1)^\top$ and covariance matrix $\begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix}$, and let $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$ with $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $\mathbf{b} = (0, 1)^\top$. Compute the mean and the covariance matrix of $\mathbf{Y}$.

**Problem 5.4** ($\star$)**.** Three random variables have variance 1 each, and every pair has correlation $\rho$. For which $\rho$ is this possible? (Hint: the covariance matrix must be positive semidefinite; find its eigenvalues.) Give an example with $\rho = -\frac{1}{2}$.

## 6 The Gaussian Distribution

**Problem 6.1** (Mandatory)**.** Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ with $\boldsymbol{\mu} = (1, 2)^\top$ and $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$.

(a) Find the distribution of $X_1 + X_2$, and $P(X_1 + X_2 > 3)$.

(b) Find the conditional distribution of $X_2$ given $X_1 = 3$, and of $X_1$ given $X_2 = 0$.

**Problem 6.2** (Mandatory). Find the lower triangular matrix $L$ with positive diagonal entries such that $LL^\top = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$ (the Cholesky factorization). Which sample of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ from Problem 6.1 does the standard normal vector $\mathbf{z} = (1, -1)^\top$ produce?

**Problem 6.3.** Let $X \sim \mathcal{N}(0, 1)$ be a signal, and let $Y = 2X + \varepsilon$ be a noisy measurement with independent noise $\varepsilon \sim \mathcal{N}(0, 1)$. Find the joint distribution of $(X, Y)$, and the conditional distribution of $X$ given $Y = y$. What is the best estimate of the signal from the measurement $y = 5$?

**Problem 6.4.** Let $X \sim \mathcal{N}(1, 4)$ and $Y \sim \mathcal{N}(-1, 9)$ be independent. Find the distribution of $X - Y$, and $P(X > Y)$.

**Problem 6.5** ($\star$). Let $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I)$ in $\mathbb{R}^d$. Show that $\mathbb{E}[\|\mathbf{Z}\|^2] = d$ and $\mathrm{Var}(\|\mathbf{Z}\|^2) = 2d$ (use $\mathbb{E}[Z_i^4] = 3$). For $d = 100$, between which values does $\|\mathbf{Z}\|$ typically lie? Although the density is largest at $\mathbf{0}$, almost no samples are close to $\mathbf{0}$; explain.

# 7 Limit Theorems and Monte Carlo Methods

**Problem 7.1** (Mandatory). A fair die is rolled 600 times.

(a) Find the mean and the standard deviation of the sum $S$ of the results, and use the central limit theorem to approximate $P(S \geq 2200)$.

(b) Approximate the probability of at least 120 sixes.

**Problem 7.2** (Mandatory). The click rate $p$ of an online ad is about 0.05. How many users must see the ad so that the observed click rate is within $\pm 0.005$ of $p$ with probability 95%? Answer with the central limit theorem, and compare with the guarantees from Chebyshev's inequality (with $\sigma^2 = 0.05 \cdot 0.95$) and Hoeffding's inequality.

**Problem 7.3.** Explain how to estimate $\int_0^1 e^{-x^2}\,dx$ by Monte Carlo with uniform random numbers. The integral is 0.7468, and $\int_0^1 e^{-2x^2}\,dx \approx 0.5981$. What is the standard error of the estimate with $N = 10\,000$ samples?

**Problem 7.4** ($\star$). The law of large numbers needs its assumptions.

(a) Let $X_1 = X_2 = \cdots = X$ for a single random variable $X \sim \mathrm{Bernoulli}(\frac{1}{2})$. Then the $X_i$ are identically distributed but not independent. What is the distribution of $\bar{X}_N$? Does it converge to $\frac{1}{2}$?

(b) The Cauchy distribution (Problem 2.5) has no mean, since $\int |x| \frac{1}{\pi(1+x^2)}\,dx = \infty$. One can show that the mean of $N$ i.i.d. Cauchy random variables is again Cauchy distributed. What does this mean for the law of large numbers?

**Problem 7.5** ($\star$). Let $X_1, \ldots, X_N$ be i.i.d. with mean $\mu \neq 0$ and variance $\sigma^2$. Use the delta method to show that $\sqrt{N}(\bar{X}_N^2 - \mu^2)$ converges in distribution to $\mathcal{N}(0, 4\mu^2\sigma^2)$. For $\mu = 2$, $\sigma^2 = 1$ and $N = 100$, what is the approximate standard error of $\bar{X}_N^2$ as an estimate of $\mu^2 = 4$? What changes for $\mu = 0$?

**Problem 7.6** (Mandatory). (a) To sample from the density $p(x) = 2x$ on $[0, 1]$ by rejection sampling with the uniform proposal on $[0, 1]$, find the smallest $M$ with $p \leq Mq$, and the acceptance rate. Describe the algorithm in words.

(b) Do the same for the Beta$(2,2)$ density $6x(1-x)$.

(c) For the target $\mathcal{N}(\mathbf{0}, I)$ and the proposal $\mathcal{N}(\mathbf{0}, 2I)$ in $\mathbb{R}^d$, show that the smallest $M$ is $2^{d/2}$, and compute the acceptance rate for $d = 2$, 10 and 20.

**Problem 7.7** ($\star$)**.** Let $X \sim$ Exponential(1), and let $p = P(X > 5) = e^{-5} \approx 0.0067$. We estimate $p$ by importance sampling with the proposal Exponential(0.2), which has the density $0.2e^{-0.2x}$.

(a) Find the importance weights $w(x)$, and write down the estimator.

(b) Compute the variance of a single term of the estimator, and compare it with the variance $p(1-p)$ of a single term of plain Monte Carlo. What are the relative standard errors of the two methods for $N = 10\,000$ samples?

**Problem 7.8** (**Mandatory**)**.** SGD for $f(\theta) = \frac{1}{2N}\sum_{i=1}^{N}(\theta - y_i)^2$ uses $g_k = \theta_k - y_{I_k}$ with a uniformly random index $I_k$.

(a) Show that $g_k$ is unbiased: $\mathbb{E}[g_k \mid \theta_k] = f'(\theta_k)$.

(b) With a constant step size $\eta \in (0,1)$, show that $e_k = \theta_k - \bar{y}$ satisfies $\mathbb{E}[e_{k+1}^2] = (1-\eta)^2\mathbb{E}[e_k^2] + \eta^2 s^2$, where $s^2 = \frac{1}{N}\sum_i (y_i - \bar{y})^2$, and find the limit of $\mathbb{E}[e_k^2]$.

(c) For the data $1, 2, 6$ and $\eta = 0.1$, compute this limit. How does it change if $\eta$ is halved?

# 8 Markov Chains

**Problem 8.1** (**Mandatory**)**.** Consider the Markov chain with the states $1, 2, 3$ and the transition matrix

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0.5 & 0.5 \end{pmatrix}.$$

(a) Draw the transition graph. Is the chain irreducible and aperiodic?

(b) Compute the distribution of $X_2$ if $X_0 = 1$.

(c) Find the stationary distribution, and check detailed balance.

(d) Find the eigenvalues of $P$. How fast does the distribution converge?

**Problem 8.2** (**Mandatory**)**.** A two-state chain has $P = \begin{pmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{pmatrix}$. Find the stationary distribution, the second eigenvalue of $P$, and a formula for $P(X_t = 1)$ if $X_0 = 1$. Evaluate it for $t = 3$.

**Problem 8.3.** A walker moves on the graph with the vertices $1, 2, 3, 4$ and the edges $\{1,2\}$, $\{2,3\}$, $\{3,4\}$, $\{4,1\}$, $\{1,3\}$ (a square with one diagonal), each time to a uniformly random neighbor. Write down the transition matrix, find the stationary distribution, and explain why the walk converges to it from every starting vertex, although the walk on the square without the diagonal does not.

**Problem 8.4** ($\star$)**.** (Gambler's ruin.) A gambler starts with 1 euro and bets 1 euro at a time on fair coin flips, until reaching 0 or 3 euros. Let $h_i$ be the probability of reaching 3 euros starting from $i$ euros. Explain why $h_0 = 0$, $h_3 = 1$ and $h_i = \frac{1}{2}h_{i-1} + \frac{1}{2}h_{i+1}$ for $i = 1, 2$, and solve these equations. Which states of the chain are absorbing, and why does the chain have more than one stationary distribution?

**Problem 8.5.** Construct the Metropolis chain for the target distribution $\boldsymbol{\pi} = (0.2, 0.3, 0.5)$ on the states $1, 2, 3$, with the proposal that picks one of the two other states uniformly at random. Write down the transition matrix, and check that $\boldsymbol{\pi} P = \boldsymbol{\pi}$.

**Problem 8.6.** Estimate the transition probabilities of a bigram model from the sequence "a b a a b c a b" by counting (maximum likelihood). Which transitions get probability 0, and why is this a problem for a language model?

**Problem 8.7.** Let $(X_1, X_2) \in \{0,1\}^2$ have the distribution $\pi(0,0) = \pi(1,1) = 0.4$ and $\pi(0,1) = \pi(1,0) = 0.1$.

(a) Find the conditional distributions of $X_1$ given $X_2 = 0$ and given $X_2 = 1$.

(b) Write down the $4 \times 4$ transition matrix of the Gibbs update of $X_1$ (with the states ordered $(0,0), (0,1), (1,0), (1,1)$), and check detailed balance with respect to $\pi$.

(c) Explain why the Gibbs sampler mixes slowly if $\pi(0,0) = \pi(1,1) = 0.49$ and $\pi(0,1) = \pi(1,0) = 0.01$.

**Problem 8.8** (**Mandatory**)**.** Give the chain of Problem 8.1 the rewards $\mathbf{r} = (0, 0, 1)^\top$ (only state 3 pays) and the discount factor $\gamma = \frac{1}{2}$.

(a) Write down the Bellman equation, and solve it for the value vector $\mathbf{V}$.

(b) Carry out two steps of iterative policy evaluation, starting from $\mathbf{V}_0 = \mathbf{0}$. Compare the error $\|\mathbf{V}_2 - \mathbf{V}\|_\infty$ with the bound $\gamma^2 \|\mathbf{V}_0 - \mathbf{V}\|_\infty$.

(c) Why is $V_1 < V_2 < V_3$?

**Problem 8.9.** A Markov decision process has the states 1 and 2, the actions "stay" and "move", and the discount factor $\gamma = 0.9$. In state 1, "stay" gives the reward 0 and stays in state 1, and "move" costs 1 (reward $-1$) and leads to state 2. In state 2, "stay" gives the reward 1 and stays in state 2 with probability 0.8 (otherwise it leads to state 1), and "move" gives the reward 0 and leads to state 1.

(a) Carry out three steps of value iteration, starting from $\mathbf{V}_0 = \mathbf{0}$.

(b) Compute the values of the policy that always stays, and of the policy that moves in state 1 and stays in state 2, by solving their Bellman equations.

(c) Check that the second policy is greedy with respect to its own value vector, i.e., that its value satisfies the Bellman optimality equation. What is the optimal policy, and why is it worth paying for the move?

**Problem 8.10** (**Mandatory**)**.** Let $X_t = 0.5 X_{t-1} + \varepsilon_t$ with white noise $\varepsilon_t$ of variance $\sigma^2 = 3$.

(a) Find the variance and the autocovariances $c(h)$ of the stationary process, and $\rho(2)$.

(b) Let the noise be Gaussian. Given $X_t = 2$, find the best prediction of $X_{t+2}$ and the variance of its error.

(c) Is $X_t = 1.1 X_{t-1} + \varepsilon_t$ stationary? Why not?

(d) Estimate $\phi$ by least squares from the observed series $1, 2, 0, -1, 1$.

## 9 Parameter Estimation

**Problem 9.1** (Mandatory)**.** Compute the maximum likelihood estimates for the following data, assuming i.i.d. samples:

(a) lifetimes $2, 5, 1, 4, 3$ from an Exponential($\lambda$) distribution;

(b) counts $2, 0, 3, 1, 4$ from a Poisson($\lambda$) distribution;

(c) measurements $4, 7, 5, 8, 6$ from $\mathcal{N}(\mu, \sigma^2)$; also compute the unbiased estimate $s^2$.

**Problem 9.2** (Mandatory)**.** Let $X_1, \dots, X_N$ be i.i.d. with mean $\mu$ and variance $\sigma^2$, and consider the shrinkage estimators $c\bar{X}_N$ for $c \in \mathbb{R}$.

(a) Show that the MSE of $c\bar{X}_N$ is $(c-1)^2\mu^2 + c^2\frac{\sigma^2}{N}$, and find the $c$ that minimizes it.

(b) For $\mu = 1$, $\sigma^2 = 4$ and $N = 4$, compare the MSE of the best $c\bar{X}_N$ with the MSE of $\bar{X}_N$. Why can we not use this estimator in practice, and what does it suggest?

**Problem 9.3** (Mandatory)**.** Use the method of moments to estimate:

(a) the shape $k$ and the scale $s$ of a Gamma distribution (mean $ks$, variance $ks^2$), for data with the sample mean $4$ and the sample variance $8$;

(b) $\theta$ for Uniform$(0, \theta)$ data, using the serial numbers $19, 40, 42, 60$ of Problem 9.6. Compare with the MLE and the unbiased estimate of that problem. Give an example of data for which the method of moments estimate is impossible;

(c) $\lambda$ for Poisson($\lambda$) data, and compare with the MLE.

**Problem 9.4** (Mandatory)**.** (Fisher information.)

(a) For Bernoulli($\theta$), compute the score $s(x;\theta)$. Check that $\mathbb{E}[s] = 0$, and that both formulas $\mathbb{E}[s^2]$ and $-\mathbb{E}[\frac{\partial s}{\partial \theta}]$ give $I(\theta) = \frac{1}{\theta(1-\theta)}$.

(b) Compute the Fisher information of Poisson($\lambda$) and the Cramér–Rao bound for $\lambda = 3$ and $N = 50$. Which estimator attains it?

(c) In Problem 9.7, compute the standard error of $\hat{p}$ from the observed information $-\ell''(\hat{p})$, and compare.

**Problem 9.5.** Fit the model $y = w_0 + w_1 x + \varepsilon$ with Gaussian noise to the data $(0, 1)$, $(1, 3)$, $(2, 4)$, $(3, 8)$ by maximum likelihood: find $\hat{w}_0$, $\hat{w}_1$ and the MLE of the noise variance.

**Problem 9.6** ($\star$)**.** (The German tank problem.) Tanks are numbered $1, 2, \dots, \theta$, and we observe the serial numbers $19, 40, 42, 60$ of four tanks, which we assume to be a uniformly random sample without replacement. Show that the MLE of $\theta$ is $60$. One can show that $M(1 + \frac{1}{N}) - 1$, where $M$ is the largest observed number, is unbiased. Compute it. Why is the MLE biased?

**Problem 9.7.** Of 400 users who saw an ad, 72 clicked on it. Compute the estimated click rate, its standard error, and an approximate 95% confidence interval.

**Problem 9.8.** Data $x_1, \dots, x_N \in \{1, \dots, K\}$ are i.i.d. Categorical($\mathbf{p}$), and $n_k$ is the number of data points equal to $k$. Use a Lagrange multiplier (Lecture 3) to show that the MLE is $\hat{p}_k = \frac{n_k}{N}$.

**Problem 9.9.** Let $X_1, \dots, X_N$ be i.i.d. with mean $\mu$ and variance $\sigma^2 > 0$. Which of the following estimators of $\mu$ are unbiased, and which are consistent?

(a) $X_1$;

(b) $\bar{X}_N + \frac{1}{N}$;

(c) $\frac{1}{N-1}\sum_{i=1}^{N} X_i$.

Is the MLE $\max_i X_i$ of $\theta$ for Uniform$(0,\theta)$ data consistent? (Its expectation is $\frac{N}{N+1}\theta$, and its variance is $\frac{N\theta^2}{(N+1)^2(N+2)}$.)

**Problem 9.10** ($\star$)**.** Let $X_1, \dots, X_N$ be i.i.d. $\mathcal{N}(\mu, v)$ with known $\mu$, and consider the variance $v = \sigma^2$ as the parameter.

(a) Show that $I(v) = \frac{1}{2v^2}$.

(b) Show that $\hat{v} = \frac{1}{N}\sum_i (X_i - \mu)^2$ is unbiased with the variance $\frac{2v^2}{N}$, so that it is efficient. (Use $\mathbb{E}[Z^4] = 3$ for $Z \sim \mathcal{N}(0,1)$.)

(c) If $\mu$ is unknown, the sample variance $s^2$ of Gaussian data has the variance $\frac{2v^2}{N-1}$. Compare with the bound. What is the price of not knowing $\mu$?

**Problem 9.11** ($\star$)**.** Let $X_1, \dots, X_N$ be i.i.d. Bernoulli$(\theta)$ with $N \geq 2$, and $S = X_1 + \cdots + X_N$. We want to estimate $\theta^2$, the probability that two independent trials both succeed.

(a) Show that $X_1 X_2$ is an unbiased estimator of $\theta^2$ with the variance $\theta^2(1-\theta^2)$.

(b) Show that $\mathbb{E}[X_1 X_2 \mid S] = \frac{S(S-1)}{N(N-1)}$. (Hint: given $S = s$, the positions of the $s$ ones are uniformly random.)

(c) For $\theta = 0.3$ and $N = 8$, the variance of the estimator of (b) is about 0.011. Compare with (a).

**Problem 9.12 (Mandatory).** (a) Write the Geometric$(p)$ pmf $p(1-p)^{x-1}$, $x = 1, 2, \dots$, in the form $h(x)\exp(\eta T(x) - A(\eta))$.

(b) Compute $A'(\eta)$ and check that it is $\mathbb{E}[X] = \frac{1}{p}$.

(c) Find the MLE of $p$ from the data $x_1, \dots, x_N$ by moment matching.

**Problem 9.13.** Use the factorization criterion to find a sufficient statistic for i.i.d. data from (a) Exponential$(\lambda)$, (b) Uniform$(0,\theta)$, and (c) Geometric$(p)$.

**Problem 9.14 (Mandatory).** (a) Write the Exponential$(\lambda)$ density $\lambda e^{-\lambda x}$ $(x \geq 0)$ in the form $h(x)\exp(\eta T(x) - A(\eta))$, and find $\eta$, $T$ and $A$. Check that $A'(\eta)$ and $A''(\eta)$ are the mean and the variance, and find the MLE of $\lambda$ from $A'(\hat{\eta}) = \bar{x}$.

(b) A Poisson regression model with the features $(1, x)$ predicts counts $y$ with $\mathbb{E}[y \mid x] = e^{w_0 + w_1 x}$. For the data $(x, y) = (0,1)$, $(1,2)$, $(2,4)$, compute the gradient of the negative log-likelihood at $\mathbf{w} = \mathbf{0}$, and show that $\mathbf{w} = (0, \log 2)$ is the MLE.

**Problem 9.15.** Consider the data $1, 2, 2, 4, 6$.

(a) Draw the empirical CDF $\hat{F}_5$, and give $\hat{F}_5(2)$ and $\hat{F}_5(5)$.

(b) Compute the kernel density estimate with the Gaussian kernel and $h = 1$ at $x = 2$.

(c) How many samples does the theorem on the uniform convergence of the empirical CDF need to guarantee $\sup_t |\hat{F}_N(t) - F(t)| \le 0.05$ with probability at least 0.95 (for a continuous $F$)? How many does the Dvoretzky–Kiefer–Wolfowitz bound $2e^{-2N\varepsilon^2}$ need?

**Problem 9.16.** The lifetime $T$ of a component (in years) has the Weibull distribution with the shape $k = 2$ and the scale $s = 10$, i.e., $S(t) = e^{-(t/10)^2}$.

(a) Find the hazard rate $h(t)$. Is the component subject to wear-out?

(b) Find $P(T > 10)$, and the median lifetime.

(c) Find $P(T > 15 \mid T > 10)$, and compare with $P(T > 5)$. Which value would an exponential lifetime give?

**Problem 9.17.** Five machines are observed until they fail or until the study ends. The times (in years) are $1, 2^+, 3, 4^+, 5$, where $^+$ marks a censored time.

(a) Assuming exponential lifetimes, find the maximum likelihood estimate of the failure rate $\lambda$ and of the mean lifetime. What would you get if you dropped the censored machines?

(b) Compute the Kaplan–Meier estimate $\hat{S}(t)$ for $0 \le t \le 5$.

# 10 Hypothesis Testing

**Problem 10.1** (**Mandatory**)**.** A coin shows 15 heads in 20 flips.

(a) Compute the one-sided p-value $P(S \ge 15)$ for $H_0$: the coin is fair, against $H_1$: it favors heads, and the two-sided p-value.

(b) Which of the two tests rejects $H_0$ at the level 0.05? At the level 0.01?

(c) What is the exact probability of a type I error of the test "reject $H_0$ if $S \ge 15$"?

**Problem 10.2.** A new classifier makes 160 errors on 2000 independent test examples, while the old classifier is known to have the error rate 0.10. Test $H_0$: the new error rate is 0.10 against $H_1$: it is smaller, with a z-test. Compute the p-value and a 95% confidence interval for the new error rate.

**Problem 10.3** (**Mandatory**)**.** In a genetics experiment, a cross is expected to produce dominant and recessive types in the ratio $3:1$. Among 400 offspring, 290 are dominant and 110 recessive. Compute Pearson's $\chi^2$ statistic and the approximate p-value, using $P(\chi^2_1 \ge c) = 2(1 - \Phi(\sqrt{c}))$ for one degree of freedom (why does this formula hold?). Is the ratio $3:1$ rejected at the level 0.05?

**Problem 10.4.** (a) For the one-sided z-test with known $\sigma$, show directly that the p-value $1 - \Phi(Z)$ is uniformly distributed if $H_0$ is true. How are the p-values distributed if $H_1$ is true?

(b) A scientist tests 100 hypotheses that are all true at the level 0.05. How many rejections do you expect?

**Problem 10.5** (Mandatory). Eight tests give the p-values

$$0.002,\ 0.009,\ 0.013,\ 0.021,\ 0.04,\ 0.12,\ 0.3,\ 0.7.$$

(a) How many hypotheses are rejected at the level 0.05 without correction? If all null hypotheses were true and the tests independent, what would be the probability of at least one rejection?

(b) How many are rejected with the Bonferroni correction at $\alpha = 0.05$?

(c) How many are rejected by the Benjamini–Hochberg procedure with $q = 0.05$?

**Problem 10.6** (Mandatory). Group A has the values $8, 11, 12$, and group B the values $5, 6, 7, 9$. Compute the difference of the group means, and the exact one-sided p-value of the permutation test with this statistic. (Hint: the difference only depends on the sum of the three values assigned to group A.)

**Problem 10.7.** For the data $2, 4, 9$:

(a) How many different bootstrap samples are there, if the order of the values does not matter?

(b) Compute the bootstrap standard error of the mean (for $B \to \infty$), and compare it with $\frac{s}{\sqrt{N}}$.

**Problem 10.8** ($\star$). Data $x_1, \dots, x_N$ are i.i.d. $\mathcal{N}(\mu, 1)$. Consider the one-sided z-test of $H_0$: $\mu = 0$ against $H_1$: $\mu > 0$ at the level 0.05 (with $z_{0.95} \approx 1.645$).

(a) For $N = 25$, for which values of $\bar{x}$ is $H_0$ rejected? What is the power of the test if $\mu = 0.5$?

(b) How large must $N$ be for a power of 0.9 if $\mu = 0.5$ (with $z_{0.9} \approx 1.2816$)?

**Problem 10.9** (Mandatory). A single observation $X$ is Exponential($\lambda$). We test $H_0$: $\lambda = 1$ against $H_1$: $\lambda = 2$.

(a) Compute the likelihood ratio $\frac{p_1(x)}{p_0(x)}$, and show that the likelihood ratio tests reject $H_0$ for small $x$.

(b) Find the rejection region of the most powerful test at the level 0.05, and its power.

**Problem 10.10** ($\star$). A coin is flipped 10 times. Find the most powerful test of $H_0$: $p = 0.5$ against $H_1$: $p = 0.7$ whose level is at most 0.05 among the likelihood ratio tests, its exact level and its power. Why is the level smaller than 0.05?

**Problem 10.11** (Mandatory). Consider the data of Problem 9.5 and the model $y = w_0 + w_1 x + \varepsilon$ with i.i.d. $\mathcal{N}(0, \sigma^2)$ noise.

(a) Compute the unbiased estimate $s^2$ of $\sigma^2$ and the standard error of $\hat{w}_1$. (Use $\mathrm{Var}(\hat{w}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$.)

(b) Test $H_0$: $w_1 = 0$ at the level 0.05, using $t_{2,0.975} = 4.30$.

(c) Give a 95% confidence interval for $w_1$. Why is it so wide, although the slope is clearly positive?

**Problem 10.12.** In the model $y_i = w x_i + \varepsilon_i$ with $x = 1, 2, 3, 4$, the noise has the standard deviations $\sigma_i = \sigma x_i$.

(a) Compute the variance of the least squares estimate $\hat{w} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}$, and compare it with the variance $\frac{\sigma^2}{N}$ that the usual formula gives with the average noise variance $\bar{\sigma}^2 = \frac{\sigma^2}{N} \sum_i x_i^2$. By which factor is the usual standard error too small?

(b) Show that the weighted least squares estimate $\tilde{w} = \frac{1}{4} \sum_i \frac{y_i}{x_i}$ is unbiased, and compare its variance with that of $\hat{w}$.

## 11 Evaluating Models and Experiments

**Problem 11.1** (**Mandatory**)**.** Two models are evaluated with 5-fold cross-validation on the same folds. Their fold errors are $0.20, 0.18, 0.25, 0.22, 0.20$ (model A) and $0.18, 0.17, 0.22, 0.21, 0.18$ (model B).

(a) Compute the cross-validation error, the standard deviation of the fold errors, and the naive standard error of model A. Why is the naive standard error too optimistic?

(b) Compute the paired differences $A - B$, their mean and standard deviation, and the statistic $\frac{\bar{d}}{s_d/\sqrt{5}}$. Compare with the statistic that treats the two models' errors as independent. Why is the paired comparison so much sharper?

**Problem 11.2** ($\star$)**.** Ten hyperparameter settings all have the true error 0.20, and their cross-validation estimates have independent errors with standard deviation 0.02. The expected maximum of 10 independent standard normal random variables is about 1.54. What is the expected cross-validation error of the selected (best-looking) setting? How should its error be estimated instead?

**Problem 11.3** (**Mandatory**)**.** In an A/B test, 120 of 2400 users convert with version A and 150 of 2400 with version B.

(a) Compute the two-proportion z-statistic and the two-sided p-value. Is the difference significant at the level 0.05?

(b) Compute a 95% confidence interval for $p_B - p_A$.

(c) How many users per group would be needed to detect an increase from 5% to 6.25% with power 0.8 at the level 0.05?

**Problem 11.4** ($\star$)**.** An A/B test with two identical versions is analyzed twice: after half of the users, and at the end. The two z-statistics $Z_1$ and $Z_2$ are approximately standard normal, with correlation $\sqrt{1/2}$ (they share half of the data). Explain why $P(|Z_1| > 1.96 \text{ or } |Z_2| > 1.96) > 0.05$; a simulation gives about 0.083. How can you keep the false positive rate at 0.05 with two looks?

**Problem 11.5.** Two models are evaluated on test examples, and their losses on a single example have variance 1 each and correlation 0.8. Compare the variance of the estimated difference of the mean losses when both models are evaluated on the same 1000 examples, and when each model is evaluated on its own 1000 independent examples. How many examples would the second design need for the same precision?

**Problem 11.6** (**Mandatory**)**.** A screening classifier is applied to 1000 patients, of whom 20 have a disease. It flags 50 patients, of whom 15 have the disease.

(a) Write down the confusion matrix, and compute the accuracy, precision, recall, false positive rate and $F_1$ score.

(b) Is the accuracy a good summary here? What accuracy does the classifier that flags nobody have?

**Problem 11.7.** Three positive examples have the scores 0.9, 0.6, 0.35, and four negative examples have the scores 0.8, 0.5, 0.3, 0.1. Draw the ROC curve, and compute the AUC as an area and as a fraction of pairs.

**Problem 11.8.** How large must a test set be to estimate the error rate of one classifier within $\pm 0.01$ with probability at least 0.95, by Hoeffding's inequality? How large must it be for 1000 classifiers simultaneously? What happens if the same test set is used to select the best of the 1000 classifiers and to report its error?

**Problem 11.9.** Two models are evaluated on easy and hard queries. Model A answers 95 of 100 easy and 120 of 300 hard queries correctly, and model B 280 of 300 easy and 35 of 100 hard queries.

(a) Compute the accuracies of both models on the easy queries, on the hard queries, and overall. Which model looks better?

(b) Explain the paradox, and compute the accuracies of both models for a query mix with 50% easy and 50% hard queries.

(c) How should the evaluation have been designed to avoid the problem?

## 12 Bayesian Inference

**Problem 12.1** (**Mandatory**)**.** With the prior $\mathrm{Beta}(3, 3)$, a coin shows 9 heads in 10 flips. Find the posterior distribution, the posterior mean, the MAP estimate, the MLE, and the predictive probability that the next flip shows heads. Explain the differences.

**Problem 12.2** (**Mandatory**)**.** Measurements $x_i \sim \mathcal{N}(\mu, 4)$ are i.i.d., and the prior is $\mu \sim \mathcal{N}(0, 1)$. Four measurements have the mean $\bar{x} = 3$. Find the posterior distribution of $\mu$ (use the formula of the conjugacy exercise in the lecture). How do the posterior mean and variance change with 100 measurements with the same mean?

**Problem 12.3.** The number of goals of a football team per match is modeled as $\mathrm{Poisson}(\lambda)$, with the prior $\mathrm{Gamma}(3, 2)$ (shape 3 and rate 2, i.e., the density $\propto \lambda^2 e^{-2\lambda}$, with the mean 1.5). In four matches, the team scores 1, 0, 2 and 1 goals.

(a) Find the posterior distribution, its mean, and the MLE of $\lambda$.

(b) Compute the predictive probability $\int_0^\infty e^{-\lambda} p(\lambda \mid \mathcal{D})\, d\lambda$ that the team scores no goal in the next match, and compare it with $e^{-\hat{\lambda}}$ for the MLE $\hat{\lambda}$.

**Problem 12.4** (**Mandatory**)**.** Let $f \sim \mathcal{GP}(0, k)$ with the squared exponential kernel $k(x, x') = \exp\big(-\frac{(x-x')^2}{2}\big)$ (the length scale $\ell = 1$), and suppose that we observe $y = 1$ at $x = 0$, with the noise variance $\sigma^2 = 0.25$.

(a) Show that the posterior mean and variance of $f(x_*)$ are $0.8\, e^{-x_*^2/2}$ and $1 - 0.8\, e^{-x_*^2}$.

(b) Evaluate them at $x_* = 0, 1, 2, 3$, and sketch them. Explain the behavior far away from the data.

(c) What happens at $x_* = 0$ as $\sigma^2 \to 0$? Why is the posterior mean at 0 smaller than the observed value?

**Problem 12.5.** (MAP with a Laplace prior.) Show that the minimizer of $f(w) = \frac{1}{2}(w-a)^2 + \lambda|w|$ for $\lambda > 0$ is the soft-thresholding $w^* = \mathrm{sign}(a)\max(|a| - \lambda, 0)$. (Hint: consider $w > 0$, $w < 0$ and $w = 0$ separately.) Explain why $\ell_1$ regularization produces exactly zero weights.

**Problem 12.6.** Fit the model $y = wx + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.5)$ and the prior $w \sim \mathcal{N}(0, 1)$ to the data $(1, 1)$, $(2, 3)$ with Bayesian linear regression. Find the posterior of $w$, and the predictive distribution of $y$ at $x = 3$. Check that the posterior mean is the ridge solution with $\lambda = \alpha\sigma^2$.

**Problem 12.7.** (Valid kernels.) A function $k$ is a valid covariance function if all matrices with the entries $K_{ij} = k(x_i, x_j)$ are symmetric positive semidefinite.

(a) Show that $k(x, x') = 1 + xx'$ and $k(x, x') = \cos(x - x')$ are valid by finding feature maps, and describe the random functions of the two Gaussian processes. (Hint: $\cos(a - b) = \cos a \cos b + \sin a \sin b$.)

(b) Show that the sum of two valid kernels is valid.

(c) Show that $k(x, x') = (x - x')^2$ is not valid. (Hint: take two points.)

(d) For the squared exponential kernel with the length scale $\ell$, compute the correlation of $f(0)$ and $f(x)$ for $x = \ell, 2\ell, 3\ell$.

**Problem 12.8** ($\star$)**.** In the medical test example of the lecture (prevalence 1%, sensitivity 99%, false positive rate 5%), a person tests positive twice, with independent tests. Use the posterior after the first test as the prior for the second test to find the probability that the person is sick. Check the result with a single application of Bayes' theorem.

**Problem 12.9** ($\star$)**.** (Choosing a hyperparameter by the marginal likelihood.) Let $f \sim \mathcal{GP}(0, c\,k)$ with $k(x, x) = 1$ and an unknown scale $c > 0$, and suppose that we observe a single value $y = f(x) + \varepsilon$ with the noise variance $\sigma^2$. Show that $y \sim \mathcal{N}(0, c + \sigma^2)$, and that the marginal likelihood $p(y)$ is maximized by $c = y^2 - \sigma^2$ if $y^2 > \sigma^2$, while it increases as $c \to 0$ if $y^2 \le \sigma^2$. Evaluate for $y = 2$ and $\sigma^2 = 0.5$, and interpret.

# 13 Information Theory

**Problem 13.1** (**Mandatory**)**.** Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$.

(a) Compute $H(p)$ in bits, and compare with the entropy of the uniform distribution on four outcomes. Compute $\mathrm{KL}(p \,\|\, u)$.

(b) Encode the four outcomes with the codewords 0, 10, 110, 111. Compute the average codeword length under $p$, and compare it with $H(p)$.

**Problem 13.2** (**Mandatory**)**.** A classifier predicts the class probabilities $q = (0.2, 0.7, 0.1)$.

(a) Compute the cross-entropy loss if the correct class is 2.

(b) With the "soft label" $p = (0.1, 0.8, 0.1)$, compute $H(p, q)$, $H(p)$ and $\mathrm{KL}(p \,\|\, q)$, and check $H(p, q) = H(p) + \mathrm{KL}(p \,\|\, q)$.

**Problem 13.3.** Compute $\mathrm{KL}\left(\mathcal{N}(0, 1) \,\|\, \mathcal{N}(1, 4)\right)$ and $\mathrm{KL}\left(\mathcal{N}(1, 4) \,\|\, \mathcal{N}(0, 1)\right)$. Explain why the second is larger.

**Problem 13.4.** (a) Show that the differential entropy of the Exponential($\lambda$) distribution is $1 - \log\lambda$.

(b) Compare it with the entropy of a Gaussian with the same variance $\frac{1}{\lambda^2}$. Show that the difference does not depend on $\lambda$, and compute it. Which theorem says that it must be nonnegative?

**Problem 13.5** ($\star$)**.** A fair bit $X$ is sent over a noisy channel, which flips it with probability 0.1; let $Y$ be the received bit. Compute the mutual information $I(X;Y) = H(Y) - H(Y \mid X)$ in bits, where $H(Y \mid X) = \sum_x p_X(x) H(Y \mid X = x)$. Compare with the cases of a perfect channel and a channel that flips with probability $\frac{1}{2}$.

**Problem 13.6** ($\star$)**.** Let $E_1, \dots, E_K$ be numbers ("energies"). Use Lagrange multipliers (Lecture 3) to maximize the entropy $H(p) = -\sum_k p_k \log p_k$ over all probability vectors $p$ with a prescribed mean energy,

$$\sum_{k=1}^{K} p_k E_k = \bar{E}.$$

Show that the maximizer has the form $p_k = \frac{e^{-\beta E_k}}{\sum_j e^{-\beta E_j}}$, i.e., $p = \mathrm{softmax}(-\beta \mathbf{E})$ for some $\beta$. (This is the Boltzmann distribution of statistical physics.)

## 14 Latent Variable Models and the EM Algorithm

**Problem 14.1** (Mandatory)**.** Run Lloyd's algorithm for $k$-means with $K = 2$ on the numbers $0, 1, 3, 8, 9, 11$, starting with the centers 0 and 3. Give the assignments, the centers and the value of $J$ after each step, until nothing changes.

**Problem 14.2.** Consider a mixture of two Gaussians with variance 1 and equal weights, and the data $1, 2, 4, 5$.

(a) Starting with the means $\mu_1 = 0$ and $\mu_2 = 3$, compute the responsibilities of the first component for all four data points.

(b) Compute the new means after one M-step, and compare them with the centers found by $k$-means from the same starting point.

**Problem 14.3** ($\star$)**.** In each of 5 experiments, one of two coins with the unknown head probabilities $\theta_A$ and $\theta_B$ is chosen with probability $\frac{1}{2}$ and flipped 10 times. The numbers of heads are $5, 9, 8, 4, 7$. Starting from $\theta_A = 0.6$ and $\theta_B = 0.5$, carry out one EM step: compute the posterior probability that coin A was used in each experiment, and the new estimates $\theta_A = \frac{\sum_i r_i h_i}{10 \sum_i r_i}$ and $\theta_B$, where $h_i$ is the number of heads and $r_i$ the posterior probability of coin A. Derive this M-step from the expected complete-data log-likelihood.

**Problem 14.4** (Mandatory)**.** (a) A binary latent variable has $P(z = 1) = \frac{1}{2}$, and a binary observation has $P(x = 1 \mid z = 0) = 0.2$ and $P(x = 1 \mid z = 1) = 0.8$. We observe $x = 1$. Compute $\log p(x)$, the posterior $p(z \mid x)$, and the ELBO for $q(z = 1) = q(z = 0) = \frac{1}{2}$. Check that the gap is $\mathrm{KL}(q \,\|\, p(\cdot \mid x))$. Which $q$ maximizes the ELBO?

(b) In a variational autoencoder with the prior $\mathcal{N}(0, 1)$, the encoder outputs $q = \mathcal{N}(1, 0.5^2)$ for a data point. Compute the KL term $\mathrm{KL}(q \,\|\, \mathcal{N}(0, 1))$ of the ELBO.

(c) With the reparameterization $z = m + s\varepsilon$, $\varepsilon \sim \mathcal{N}(0, 1)$, compute $\frac{\partial}{\partial m}\mathbb{E}[z^2]$ and $\frac{\partial}{\partial s}\mathbb{E}[z^2]$ by differentiating inside the expectation, and check the results with $\mathbb{E}[z^2] = m^2 + s^2$.

**Problem 14.5.** Data in $\mathbb{R}^3$ have the sample covariance matrix $S = \begin{pmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, and we fit probabilistic PCA with $k = 1$.

(a) Find the eigenvalues and eigenvectors of $S$, and the maximum likelihood estimates $\hat{\sigma}^2$ and $\hat{W}$.

(b) Compute the fitted covariance matrix $\hat{C} = \hat{W}\hat{W}^\top + \hat{\sigma}^2 I_3$ and its eigenvalues. Which variances does the model reproduce, and which does it average?

(c) Compute the posterior mean $\mathbb{E}[z \mid \mathbf{x}]$ for a data point with $\mathbf{x} - \hat{\boldsymbol{\mu}} = (1, 1, 0)^\top$.

## 15 Stochastic Differential Equations

**Problem 15.1** (Mandatory)**.** Let $B(t)$ be a Brownian motion.

(a) Find the distribution of $B(4) - B(1)$, and $P(B(4) > 2)$.

(b) Compute $\mathrm{Corr}(B(1), B(2))$ and the distribution of $B(1) + B(2)$. (Hint: write $B(2) = B(1) + (B(2) - B(1))$.)

(c) Given $B(1) = 1$, what is the distribution of $B(2)$? Given $B(2) = 2$, what is the distribution of $B(1)$? (Use the conditioning theorem for Gaussians.)

**Problem 15.2** (Mandatory)**.** Consider the SDE $dX = -2X\,dt + 0.5\,dB$ with $X(0) = 1$.

(a) Perform two Euler–Maruyama steps with $h = 0.1$, using the standard normal samples $Z_0 = 0.5$ and $Z_1 = -1$.

(b) Find $\mathbb{E}[X_k]$ for the Euler–Maruyama chain, and compare $\mathbb{E}[X_{10}]$ with the mean $e^{-2t}$ of the SDE at $t = 1$.

(c) Find the stationary variance of the SDE, and of the chain with $h = 0.1$. For which step sizes does the chain have a stationary distribution?

**Problem 15.3.** (Why $\sqrt{h}$?) Divide $[0, t]$ into $n = t/h$ steps, and consider a scheme that adds the noise $\sigma h^\gamma Z_k$ in step $k$, with i.i.d. $Z_k \sim \mathcal{N}(0, 1)$. Compute the variance of the total noise $\sigma h^\gamma (Z_0 + \cdots + Z_{n-1})$, and show that as $h \to 0$, it tends to 0 for $\gamma > \frac{1}{2}$ and to $\infty$ for $\gamma < \frac{1}{2}$, while it equals $\sigma^2 t$ for $\gamma = \frac{1}{2}$.

**Problem 15.4** ($\star$)**.** (Quadratic variation.) Divide $[0, T]$ into $n$ intervals of the length $h = T/n$, and let

$$Q_n = \sum_{k=0}^{n-1} \big(B((k+1)h) - B(kh)\big)^2.$$

(a) Using $\mathbb{E}[Z^4] = 3$ for $Z \sim \mathcal{N}(0, 1)$, show that $\mathbb{E}[Q_n] = T$ and $\mathrm{Var}(Q_n) = 2Th$, so that $Q_n \to T$ as $n \to \infty$.

(b) Show that for a continuously differentiable function $f$ on $[0, T]$, the sum $\sum_k \big(f((k+1)h) - f(kh)\big)^2$ tends to 0. (Hint: mean value theorem.) Conclude that Brownian paths are not continuously differentiable.

(This is the rule "$(dB)^2 = dt$" of stochastic calculus.)

**Problem 15.5** (Mandatory)**.** (Langevin sampling of a Gaussian.) Let $U(x) = \frac{(x-\mu)^2}{2s^2}$, so that $e^{-U}$ is proportional to the density of $\mathcal{N}(\mu, s^2)$.

(a) Write down the unadjusted Langevin algorithm for $U$ with the step $h$, and show that it has the form $x_{k+1} - \mu = a(x_k - \mu) + \sqrt{2h}\, z_k$.

(b) Show that for $0 < h < 2s^2$, the chain has the stationary distribution $\mathcal{N}(\mu, \frac{2s^2}{2 - h/s^2})$. Evaluate the variance for $\mu = 2$, $s^2 = 0.25$ and $h = 0.01$.

(c) What does the algorithm do without the noise term? How is this related to MAP estimation?

**Problem 15.6** ($\star$)**.** (The forward process of a diffusion model.) A data point $x_0 = 3$ is gradually noised by $x_t = \sqrt{1-\beta}\, x_{t-1} + \sqrt{\beta}\, z_t$ for $t = 1, 2, \ldots$, with i.i.d. $z_t \sim \mathcal{N}(0, 1)$ and $\beta = 0.02$.

(a) Show by induction that $x_t \sim \mathcal{N}\big((1-\beta)^{t/2} x_0,\ 1 - (1-\beta)^t\big)$. Why is this useful for training, where $x_t$ is needed for random times $t$?

(b) Compute the mean and the variance for $t = 10$, 100 and 500. What is the distribution of $x_t$ for large $t$, whatever the data point?

(c) Show that the process is approximately the Euler–Maruyama scheme with $h = \beta$ for the Ornstein–Uhlenbeck SDE $dX = -\frac{1}{2} X\, dt + dB$. (Hint: $\sqrt{1-\beta} \approx 1 - \frac{\beta}{2}$.)

# 16 Measure-Theoretic Probability (Appendix)

**Problem 16.1** (**Mandatory**)**.** Let $\Omega = \{1, 2, 3, 4\}$.

(a) List the elements of the $\sigma$-algebra generated by the event $\{1, 2\}$, and of the $\sigma$-algebra generated by the two events $\{1\}$ and $\{2\}$.

(b) Let $X = \mathbf{1}_{\{1,2\}}$ and $Y(\omega) = \omega$. With respect to which of the two $\sigma$-algebras of (a) is $X$ measurable, and with respect to which is $Y$? What is $\sigma(X)$?

(c) Is the collection $\{\emptyset, \{1\}, \{2, 3, 4\}, \{1, 2\}, \{3, 4\}, \Omega\}$ a $\sigma$-algebra?

**Problem 16.2.** Compute the following limits, and justify exchanging the limit and the integral with the monotone or the dominated convergence theorem.

(a) $\lim_{n\to\infty} \int_0^\infty \frac{n}{n+x} e^{-x}\, dx$.

(b) $\lim_{n\to\infty} \int_0^1 \frac{nx}{1+n^2x^2}\, dx$. (Hint: the integrand is at most $\frac{1}{2}$.) Check your answer by computing the integral exactly.

(c) For the number $X$ shown by a fair die, compute $\mathbb{E}[X]$ with the tail sum formula $\mathbb{E}[X] = \sum_{k\geq 0} P(X > k)$.

**Problem 16.3.** Let $Z \sim \mathcal{N}(0, 1)$, and let $Y = \max(0, Z - 1)$ be the output of a ReLU unit with the bias $-1$.

(a) Compute $P(Y = 0)$. Does $Y$ have a density with respect to the Lebesgue measure? Find a measure with respect to which it has one, and the density.

(b) Show that $\mathbb{E}[Y] = \varphi(1) - (1 - \Phi(1))$ and $\mathbb{E}[Y^2] = 2(1 - \Phi(1)) - \varphi(1)$, and evaluate them and $\mathrm{Var}(Y)$ with $\varphi(1) \approx 0.2420$ and $\Phi(1) \approx 0.8413$. (Hint: $\varphi'(z) = -z\varphi(z)$.)

**Problem 16.4.** A fair die shows the number $X$, and an observer only learns whether $X$ is even. This information is described by the $\sigma$-algebra $\mathcal{G} = \{\emptyset, \{2,4,6\}, \{1,3,5\}, \Omega\}$.

(a) Compute $\mathbb{E}[X \mid \mathcal{G}]$ and $\mathbb{E}[X^2 \mid \mathcal{G}]$, and check the tower property.

(b) Compute $\mathrm{Var}(X \mid \mathcal{G}) = \mathbb{E}[X^2 \mid \mathcal{G}] - \mathbb{E}[X \mid \mathcal{G}]^2$, and check the law of total variance $\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid \mathcal{G})] + \mathrm{Var}(\mathbb{E}[X \mid \mathcal{G}])$.

(c) Show directly that among all random variables $Z = a\mathbf{1}_{\{X \text{ even}\}} + b\mathbf{1}_{\{X \text{ odd}\}}$, i.e., all predictions that only use the information $\mathcal{G}$, the conditional expectation minimizes $\mathbb{E}[(X - Z)^2]$.

**Problem 16.5** ($\star$)**.** (Double or nothing.) In a fair game, a gambler bets 1 euro. After every loss, the gambler doubles the stake, and after the first win, the gambler stops.

(a) Show that the gambler stops after finitely many rounds with probability 1, and has then won exactly 1 euro.

(b) Let $M_n$ be the gain after $n$ rounds (which stays constant once the gambler has stopped). Show that $(M_n)$ is a martingale, so that $\mathbb{E}[M_n] = 0$ for all $n$, although $M_n \to 1$ almost surely. Why does the optional stopping theorem not apply?

(c) What is the expected largest debt of the gambler, i.e., the expected loss just before the win?

**Problem 16.6.** (a) Let $A_1, A_2, \ldots$ be independent events with $P(A_n) = n^{-1.5}$. Do infinitely many of them happen, almost surely? What if $P(A_n) = \frac{1}{n \log n}$ for $n \geq 2$?

(b) Let $X_1, X_2, \ldots$ be independent with $P(X_n = n^2) = \frac{1}{n^2}$ and $X_n = 0$ otherwise. Does $X_n \to 0$ almost surely? In $L^1$?